

Chapter 9: The Process of Interaction Design

The author of this chapter touches on several topics that lead me to think of some issues that aren't directly addressed in the details of the reading. One of them, for example, is the *systems design* approach that is introduced as one of the four approaches to interaction design. The book makes a point of focusing on explaining user-centered design in particular, and only defines systems design as a context-focused approach good for complex problems, where the interaction "ecosystem" is taken holistically into account; in my opinion, the users seem to have a more balanced role in this model with everything taken into perspective, where they just "set the goals" of the system and everything else that is part of the system should be taken into account with equal importance, holistically. Although user-centered design does take many things into account, it seems to do so on many levels from the perspective of the user and not as much taking an equally balanced perspective of everything that isn't just the user.

Then, perhaps the values that go into approaching interaction design will also heavily affect what kinds of approaches (e.g. user-centered or systems) are used, as well as what issues are taken as a priority. For example, in user-centered design, the user/stakeholder is categorized into three levels: primary, secondary, and tertiary. Although the chapter talks about how stakeholders are all involved in some way with the product, the way the terminology is used to split these individuals into layers that still differ semantically in their "distance" from the design issue makes me wonder: in a user-centered approach, how much priority the non-primary users really take up in this process. There is a feeling that the primary user, being "closer" to the product, will be regarded with more importance in the design process. The chapter doesn't seem to go into too much detail on whether this is actually the case. In contrast, "holistic" systems design seems more likely to take a balanced view of all varying aspects. Again, the reason one approach may be chosen over another might be affected by the ideological priorities of the individual or organization directing the design project – for example, someone more interested in the STS (science, technology, and society) implications of a product may take more detailed interest in every aspect of the design, whereas a for-profit company prioritizes profit and therefore is more likely to be motivated to make making its direct users happy a greater priority.

Another issue brought up in the chapter that triggered thought of how variance in values and attitudes can really change many facets of how certain issues are managed in interaction design is the dilemma presented about copying for inspiration. The chapter talks about how there are various and shifting options to enforce and facilitate the concept of intellectual property, from copyrights and patents to emerging creative commons and open source development. I think this is a very interesting issue and makes people think about what kinds of feelings, thoughts, and beliefs lead people to stand on one side of the spectrum or the other – the spectrum of clinginess to intellectual property, as well as the differentiations made between an idea itself and the expression of it. There seem to be pros and cons to different approaches, such as copyright (expression) being easier to define and control but more restrictive, and patenting (idea) being more conducive to expression but a difficult, more abstract process to maneuver. But to go back to difference in personal values, what I would like to know is what kinds of factors (apart from the obvious, related to money-

making) make certain people care more about protecting the dissemination of ideas and others more motivated to advocate increased dissemination, sharing, and creativity. From my own perspective, in the bigger picture there is clearly more beneficial effect of encouraging idea sharing versus restricting the flow and usage of ideas, for what seems to be the sake of ownership – but perhaps there's other important factors apart from personal values which affect what stance people take on the spectrum of "intellectual property possession".

Lastly, the chapter's mention of expectation management led me to think of an issue which I'm not sure is or isn't considered an example of expectation management (or rather, the lack thereof). Since expectation management seems to be about not misrepresenting a product (or parts of it) as well as making sure product expectations are realistic, this made me think of the "misunderstanding autopilot" issue that appears to plague technology nowadays. Two examples are China Airlines Flight 006 and a publicized Tesla car crash incident from March this year.

In the China Airlines Flight 006 incident on February 19, 1985, the airplane lost all power at 12,500m and fell almost to ground level partially because the captain of the plane relied too heavily on autopilot, assuming that it was doing more to control the plane than it actually did (specifically, the autopilot did not control plane rudder when plane rudder control was necessary during the situation). The possible connection this incident has with expectation management, is that expectation **mismanagement** with respect to life-critical systems may result not only in "resistance or rejection" but also misunderstandings that lead to life-threatening situations. In this case, the pilot had an expectation of the autopilot that was beyond what it could actually do.

Similarly, a heavily-publicized fatal Tesla car crash in California from March 2018 resulted from a misuse of the Autopilot function, which after investigation was implied to have been a direct cause of the fatal crash. The driver was relying 100% on Autopilot even though they had a six-second period to react and a clear view of the wall they were heading towards. Tesla then communicated that Autopilot has functions that help driving and make accidents less likely, but is not meant to take complete responsibility over the car. Users are supposed to still watch the road and keep their hands on the wheel. To prevent expectation mismanagement, this kind of information has to be communicated clearly to the public so that users don't overshoot their assumptions of what an Autopilot system is responsible for.